AgentForge: Hands-on Workshop on AI Agents for Scientific Research Assistants

Nanyang Technological University

1 Purpose, Impact, Format

1.1 Purpose

The past year has witnessed a rapid acceleration in the capabilities of AI agents. Emerging platforms (e.g., Claude Code, Antigravity), new technologies like the Model Context Protocol (MCP), and accessible frameworks such as OpenClaw are actively democratizing the creation of personalized AI assistants. However, existing conferences and workshops[1] focused on Agentic AI for science remain highly technical, catering predominantly to researchers who are already entrenched in AI development. **There is an urgent need for practical upskilling** to ensure that scientists who are not AI experts can benefit from these transformative tools.

This proposed workshop bridges a critical gap by introducing agentic AI to domain scientists and researchers who have not yet integrated these tools into their daily workflows. We aim to empower attendees through an intensive, hands-on workshop where they will build and deploy their own local AI agents acting as customized research assistants. This will provide researchers with secure, privacy-preserving solutions necessary for handling sensitive scientific data and proprietary research. These agents will be designed to streamline critical scientific tasks, such as conducting comprehensive literature reviews, querying specialized databases, analyzing data using customized scripts, and leveraging publicly available tools for science.

1.2 Expected outcomes, Timeframe, and Measurement

This workshop focuses on understanding the fundamental concept and architecture of AI agents, providing the resources for attendees to build one from scratch and to leverage existing platforms. Participants will gain the skills necessary to develop flexible, domain-specific agents tailored to their unique needs.

Outcomes	**Timeframe**	**Measurement**
New skill: Working prototypes of personal AI researcher assistants	By the end of the event	• public GitHub repos by participants • presentations during the workshop
New insight: Perspective article	1-2 months after the event	• insights, challenges, questions for exploration gathered from participants by the end of the event. • perspective article uploaded to the arXiv and submitted to peer-reviewed journals
Potential new collaborations	By the end of the event	• post-event survey
Tutorial recordings	By the end of the event	• YouTube uploads

[1] Agents4Science (Oct2025), https://agents4science.stanford.edu/, ICLR workshop (Apr2025), https://iclr.cc/virtual/2025/workshop/23991

1.3 Proposed format (tentative dates: Aug 24-29, 2026)

The workshop is designed as a four-day, immersive in-person experience. Drawing inspiration from the successful interactive model of last year's Foundation Model Workshop at the University of Toronto,[2] our format prioritizes active participation and practical application via tutorials and hackathon. A proposed full schedule is attached in Appendix A.

Day 1: Foundations	**Day 2: Advanced Agent**	**Day 3: Hackathon**	**Day 4: Hackathon**
Keynote 1: Introduction to AI agent for science	Keynote 2: Application of agentic AI in science	Tutorial 5	Hackathon
Tutorial 1 & 2	Tutorial 3 & 4	Hackathon	Presentation
Team forming	Hackathon starts	Hackathon	Give award

- **Tutorial 1: From Chatbot to Agent.** Introducing the core differences between a standard Large Language Model (LLM) and an AI agent by exploring the ReAct (Reason + Act) framework. We will build a simple agent, e.g., a literature scout agent that can access tool like arXiv search function, to demonstrate the ReAct loop in practice.
- **Tutorial 2: Agentic memory** Scientific research requires deep context that our simple agent lack. This tutorial dives into memory and Retrieval-Augmented Generation (RAG) architectures, teaching participants how to give their agents access to domain-specific knowledge bases without hallucinating facts.
- **Tutorial 3: Agent on existing platforms, MCP, and Skills.** An assistant is only as good as the tools it can use. This session focuses on tool-calling and the Model Context Protocol (MCP) to standardize how agents interact with external data sources and software. We will also transition to more sophisticate agents with more abilities like file manipulation and complex "Skills" by showing examples of agents on existing platform like Claude code[3] and Antigravity[4], alongside scientific tools publicly accessible via MCP and Skills, such as ToolUniverse[5].
- **Tutorial 4: Multi-agents orchestration.** Complex scientific workflows often require specialized roles. This tutorial explores how to orchestrate multiple agents that communicate, debate, and delegate tasks to one another to achieve a broader research goal.
- **Tutorial 5: Agentic research assistant - Autonomous agent** A comprehensive, end-to-end example that integrates concepts from the previous tutorials, demonstrating how they come together to build a functional research assistant. Participants will learn how to combine these components into a cohesive system. The session will also introduce autonomous agents such as OpenClaw and NanoClaw—open-source frameworks that shift the paradigm from simple chatbots to "always-on," system-level agents—and explore how they can be applied to scientific tasks. (Optional hands-on) Participants will set up an instance on a remote virtual private server (VPS) and experiment with it firsthand.
- **Driving Domain-Specific Innovation (Group Hackathon).** Form working groups of 3–5 participants aligned by scientific discipline (e.g., bioinformatics, materials science, physics) to apply newly acquired skills. Participants bring their own tools, datasets, and research bottlenecks to collaboratively design, prompt, and deploy a custom agent tailored to domain-specific challenges—producing actionable insights and catalyzing future collaborations.
  - **Outputs**:
    - 1) a 5-minute presentation with live demo showcasing the agent successfully executing a domain-specific scientific task, and a brief (1-2 slides) architecture overview detailing the agent's tools, prompts, etc.
    - 2) shared repository containing the code and examples of successful/failed reasoning traces.
    - 3) 1-2 pages report writeup sharing insights, challenges, and/or questions for exploration gained from building agents for scientific tasks. Send it via email to the organizer.
  - **Awards**: Projects are evaluated on domain impact (how effectively it solves the scientific problem), technical execution (proper integration of tools and memory), popular votes, and other criteria to be decided.

[2] Foundation model for science (Nov2025), https://ai-for-science.org/
[3] https://claude.com/download
[4] https://antigravity.google/
[5] ToolUniverse: Build an AI Scientist from your LLM. https://aiscientist.tools/

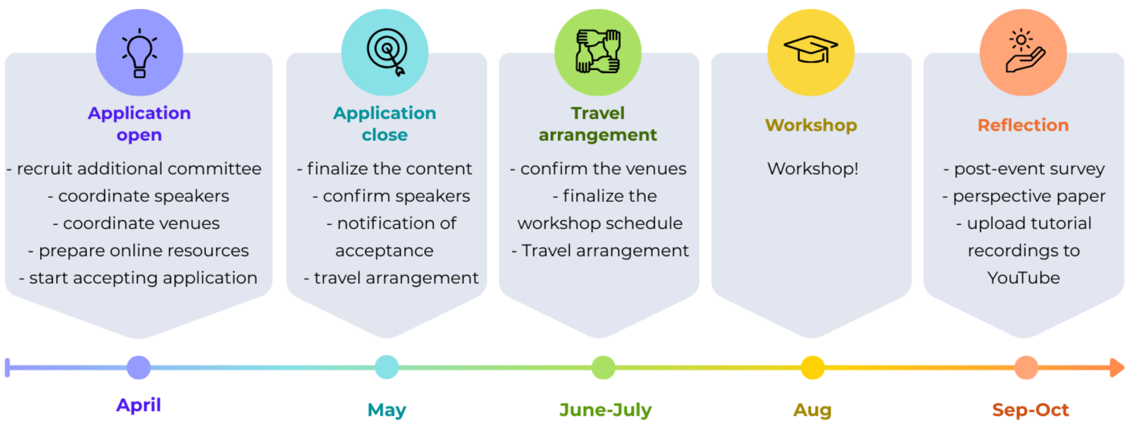
2 Logistics

2.1 Participants & Target Audience

We aim to convene approximately 35 in-person participants at NTU, representing a diverse range of disciplines spanned by Schmidt AI in Science Fellows and alumni, as well as external participants from universities. Participants will include individuals with demonstrated interest in, or relevant expertise and insights related to, the workshop themes.

There is strong and growing interest in the development of AI agents for scientific research. Building on our prior participation in the Oxford Research Software Engineering (RSE) workshop in September 2025, we collaborated on a project focused on developing an AI agent for literature discovery.[6] The core project team comprised six fellows working in person, with an additional four fellows who expressed interest and participated in selected meetings but were unable to attend on site. We are also aware of a separate proposal being submitted to this call by a group of fellows based at the University of Toronto that addresses AI agents for scientific research (Appendix B). This reflects the strong interest in this topic.

2.2 Key milestones



2.3 Key logistical elements

Location: The workshop will be held on the campus of Nanyang Technological University

Venue & Accommodation: Nanyang Executive Centre (NEC) in the campus for 4 nights.

Dining: Breakfast, lunch, and coffee breaks are usually arranged through NTU Catering that meet diverse dietary preferences and budgets. For dinners, there are restaurant suppliers with a University-Wide Agreement that accommodates large groups.

Traveling: Nanyang Technological University (NTU) in Singapore offers a strategic geographic advantage, utilizing Changi Airport's extensive global connectivity to easily attract top-tier attendees from Asia, Europe and the Americas.

2.4 Organizing team

Chengran Yang (Schmidt Fellow, School of Physical and Mathematical Sciences) - Chengran Yang works on tensor networks, quantum information, quantum algorithms, machine learning, quantum machine learning, quantum stochastic modelling and quantum finance, pushing the boundaries of applications in artificial intelligence and quantum technology.

Shichuan Sun (Schmidt Fellow, Asian School of the Environment, Nanyang Technological University): Shichuan works on computational materials science, using machine learning force fields and atomistic simulations to study mineral behavior under the extreme conditions of Earth's and planetary interiors. He has been actively building

[6] Curaitor Agent: https://github.com/ritesh001/curaitor-agent

and deploying AI agents for automating research workflows such as literature analysis and data processing, and routinely uses AI coding agents in his daily research.

Nanta Sophonrat (Schmidt Fellow (Alumni), University of Michigan, Ann Arbor, Cheminformatics): Nanta works on data-driven screening of molecules for electrochemical plastic recycling applications. She began following and learning about the development of large language models (LLMs) since she joined the Schmidt AI in Science program in 2023. She co-led teams developing LLM-based and agent-based literature curation tools during the Oxford RSE workshops in 2024 and 2025. In the past, she helped organize some events and workshops, including the conference for the Thai Student Association in Europe and winter astronomy camps for high school students.

Yukun Xue (Faculty of integrated circuit, Xidian University): Yukun is jointly trained at Tsinghua University for her Master's research. Her research focuses on hardware–software co-design, architecture-level modeling, and performance simulation for emerging computing systems. Yukun is interested in leveraging AI agents to assist in architectural exploration, automated benchmarking, and cross-layer performance analysis. She aims to explore how agentic systems can accelerate research workflows in intelligent computing system design.

Endorsement: This proposal has been endorsed by Prof. Khor Khiam Aik at Nanyang Technological University, Program Lead of the Schmidt AI in Science Fellowship, confirming that the university will provide the necessary support to execute the event

Prof Khor Khiam Aik ,
Director, Fellowships & Awards (Research Support Office)

Date

3 Appendix A – Proposed Schedule

Tentative dates: August 24–29, 2026. Day 0 is arrival day; Days 1–4 are the workshop; Day 5 is departure.

Time	Day 0 Sun ARRIVAL	Day 1 Mon	Day 2 Tue	Day 3 Wed	Day 4 Thu
8:00		Breakfast	Breakfast	Breakfast	Breakfast
9:00		Opening Remarks	**Keynote 2 (& 3):** Application of Agentic AI in Science	**Tutorial 5:** Autonomous Agents	Hackathon
9:30		**Keynote 1:** Introduction to AI Agents for Science		Tutorial 5 (cont.)	Hackathon
10:30		Coffee Break	Coffee Break	Coffee Break	Coffee Break
11:00		**Tutorial 1:** From Chatbot to Agent (ReAct) (30+10 min talk) (50 min tutorial)	**Tutorial 3:** Agents on Platforms, MCP & Skills	Hackathon: finalize plan	Hackathon: Finalize Demo
12:30		Lunch	Lunch	Lunch	Lunch
1:30	Arrivals & Check-in at NEC	**Tutorial 2:** Agentic memory	**Tutorial 4:** Multi-Agent Orchestration	Hackathon	Prepare Presentations
3:00		Coffee Break	Coffee Break	Coffee Break	Coffee Break
3:30		Poster session and Hackathon: Team formation	Hackathon: brain storming, sharing user requirements/ bottleneck/tools	Hackathon: Eval Script	**Presentations, Awards & Closing**
5:00					
6:00	**Welcome Dinner**	Dinner	Dinner	Dinner	**Farewell Dinner**
7:30	Icebreaker	Social Activity	Social Activity	Social Activity	
Night	NEC @ NTU	NEC @ NTU	NEC @ NTU	NEC @ NTU	Departures

Table 1: Proposed schedule for AgentForge. Day 0 is arrival; Days 1–2 focus on tutorials; Days 3–4 on the hackathon. Tutorials are in blue, hackathon work in orange, presentations/awards in red, keynotes/social in green.

Notes on schedule design:

- Tutorials are ordered by complexity: foundations (Day 1) ⟶ advanced topics (Day 2) ⟶ autonomous systems (Day 3 morning), ensuring participants without AI agent background can follow the progression.
- Each tutorial is 90 minutes: 40 min instruction + 50 min guided hands-on coding.

Social activities:

- Day 0: Welcome dinner and icebreaker at NEC
- Day 1: Guided evening walk at NTU campus / CleanTech Park
- Day 2: Group outing to Gardens by the Bay
- Day 3: Hawker centre dinner
- Day 4: Farewell dinner

4 Appendix B – Interest in Agentic AI for Science workshop

We are also aware of a separate proposal being submitted to this call by a group of fellows based at the University of Toronto that addresses AI agents for scientific research. We have communicated with that team to coordinate efforts and ensure that the scope, objectives, and technical approach of our proposed work are clearly distinct, complementary, and self-contained. Specifically, our proposal focuses on the development of AI agents as personal research assistants, whereas their work emphasizes the application of agentic AI in scientific research, including more advanced topics such as reinforcement learning and safety guardrails. To minimize overlap—while allowing interested participants to attend both events—we will limit participation to 25, excluding organizers and invited speakers. Moreover, the geographical locations of Canada and Singapore are expected to attract largely distinct participant groups.

5 Appendix C - More details on the sessions

We provide an outline of the content we plan to cover, designed to offer a comprehensive introduction to Agentic AI for researchers who are new to the field. This structure serves as a guideline, and the specific details can be adapted based on the speaker's interests.

5.1 Introduction and overviews of the application by keynote speakers

Keynote 1: Introduction to AI agent for science

- Explain the purpose of this workshop and why it is relevant for all members of the audience.

Keynote 2&3: Application of agentic AI in science

- Illustrating the applications of generative AI agents across different scientific disciplines.

5.2 Hands-on tutorials

For each tutorial, it includes a 30min theoretical presentation given by expert in the specific field. They will focus on the foundation side. After that, the organizing team will provide hands-on tutorial, introduce everything at coding level.

- **Tutorial 1: From Chatbot to Agent.** Introducing the core differences between a standard Large Language Model (LLM) and an AI agent by exploring the ReAct (Reason + Act) framework. We will build an agent, e.g., a literature scout agent that can access tool like arXiv search function, to demonstrate the ReAct loop in practice.
  1. What is agent (context engineering)
     1. Agent VS LLM
     2. The advantage of agent
  2. Introduce the architecture of agent
     1. ReAct (Reason + Act) framework
     2. context engineering: Memory + Tool
  3. Introduce limitation of both LLM and agent
     1. Single shot
     2. Memory and reliability
  4. Example use cases
     1. Literature agent

  Tutorial based on:
  - Minimum agent code (developed by Shichuan)
  - Other examples: langchain/langgraph, google adk
- **Tutorial 2: Agent's Memory.** Scientific research requires deep context that our simple agent lack. This tutorial dives into memory and Retrieval-Augmented Generation (RAG) architectures, teaching participants how to give their agents access to domain-specific knowledge bases without hallucinating facts.
  1. Introduce the memory system of an agent
     1. Simple text: markdown
     2. Database, .e.g., SQL, vector, graph
     3. Agentic RAG, i.e., chunk + embedding
     4. SOTA methodology
  2. Memory compression
     1. Context window size
     2. Example methods for memory compression: context-resident semantic compression, KV cache compression, parametric memory integration, policy-learned memory management

  Tutorial based on:
  - Minimum agent code (developed by Shichuan)
  - Other examples: LangChain agent; RAG: chromaDB
- **Tutorial 3: MCP and Skills.** An assistant is only as good as the tools it can use. This session focuses on tool-calling and the Model Context Protocol (MCP) to standardize how agents interact with external data sources and software. We will also transition to more sophisticate agents with more abilities like file manipulation and complex

"Skills" by showing examples of agents on existing platform like Claude code[7] and Antigravity[8], alongside scientific tools publicly accessible via MCP and Skills, such as ToolUniverse[9].

1. The foundation: mechanics of tool-calling
    1. The request-response loop
    2. Defining tool schemas
    3. Error handling
2. The model context protocol (MCP)
    1. The "USB-C" for AI agents
    2. Client-server architecture
    3. Resources vs. tools
3. Transitioning to complex "skills"
    1. File manipulation
    2. Chaining skills
4. Use MCP and skills together

Tutorial based on:
- Minimum agent code (developed by Shichuan)
- Other examples: LangChain agent for MCP; LangChain Deep Agents for skills; Agent on platform: Claude code, MCP and skills: ToolUniverse

- **Tutorial 4: Multi-agents orchestration.** Complex scientific workflows often require specialized roles. This tutorial explores how to orchestrate multiple agents that communicate, debate, and delegate tasks to one another to achieve a broader research goal.

1. Why multi-agent orchestration?
    1. Role specialization
    2. Separation of concerns
    3. The concept of "state"
2. Common multi-agent architectures
    1. Sequential / pipeline
    2. Hierarchical (manager-worker)
    3. Joint debate / collaborative

Tutorial based on:
- Minimum agent code (developed by Shichuan)
- Other examples: Agent Teams Lite, Claude Code agent team, OpenClaw

- **Tutorial 5: Agentic research assistant - Autonomous agent** a comprehensive example that integrates concepts from the previous tutorials, demonstrating how they come together to build a functional research assistant. Participants will learn how to combine these components into a cohesive system.

Tutorial based on: Minimum agent code (developed by Shichuan) or examples from external speakers

[7] https://claude.com/download
[8] https://antigravity.google/
[9] ToolUniverse: Build an AI Scientist from your LLM. https://aiscientist.tools/

6 Appendix D - Potential Speakers

6.1 Keynote speakers

Our distinguished keynote speakers, who are recognized leaders in their respective disciplines, will provide a comprehensive overview of the recent advancements and progress of agent technologies within their domains.

1. **Bo An**
   - Position: Head, Division of Artificial Intelligence, College of Computing & Data Science, NTU, Singapore
   - Expertise: Artificial intelligence, Multi-agent systems
2. **Konstantin Novoselov**
   - Position: Nobel Prize in Physics, Director of the Institute for Functional Intelligent Materials (I-FIM) and a Tan Chin Tuan Centennial Professor at NUS, Singapore
   - Expertise: AI for Materials Design
3. **Ong Yew Soon**
   - Position: President's Chair Professor of Computer Science, NTU, Singapore
   - Expertise: Artificial Intelligence, Computational Intelligence, LLM Agent, AI in Education
4. **Chunyan Miao**
   - Position: President's Chair Professor of Computer Science, NTU, Singapore
   - Expertise: Human Agent Interaction, Human Computation, Cognitive Agents

6.2 Tutorial speakers

Each tutorial begins with a 30-minute presentation by a domain expert, focusing on foundational concepts. Following the theory, the organizing team will lead a hands-on session to explore the practical, coding-level applications. All necessary Jupyter notebooks will be prepared in advance and hosted on a dedicated GitHub repository. Additionally, we will feature guest lectures from external experts specializing in the theoretical aspects of agentic AI.

1. **The organizing team**
   - Position: Most of them are Schmidt AI fellows.
   - Expertise: Agentic AI and interdisciplinary scientific domains
2. **Luu Anh Tuan**
   - Position: Associate Professor, College of Computing & Data Science, NTU, Singapore
   - Expertise: Artificial Intelligence, Multi-agent systems
3. **Han Yu**
   - Position: Associate Professor, College of Computing & Data Science, NTU, Singapore
   - Expertise: Artificial Intelligence, Multi-agent systems
4. **Xinxing Xu**
   - Position: Microsoft Research Asia (MSRA) Singapore
   - Expertise: Artificial Intelligence, Multi-agent systems, Digital Health
5. **Bijun Tang**
   - Position: Presidential postdoctoral fellow at NTU
   - Expertise: AI for Materials Science
6. **Mahesh Murag**
   - Position: Applied AI Team, Anthropic
   - Expertise: LLM, Agent

7 Appendix E - Potential audiences

Current fellows

- Guido Herrera (Cornell)
- Siddarth Achar (UChicago)
- Radha Mastandrea (UChicago)
- Alex Wikner (UChicago)
- Austin Mroz (Imperial college)
- Carla Nathaly Villacís Núñez (UM)
- Yan Ying Tan (UM)
- Xinyu Liu (UM)

Alumni

- Ritesh Kumar (UChicago, alumni)
- Taniya Kapoor (Assistant Professor at WUR, alumni)
- Caitlin Aamodt (UCSD, alumni)
- Soumi Tribedi (UM, alumni)
- Jacob Brev (UM, alumni)

External participants

- Ximing Wang (NTU)
- Naixu Guo (NUS)